

Computer Networks Lab 04

Course: Computer Networks (CL3001)

Instructor: Sadaf Zehra

Semester: Fall 2026

T.A: N/A

Note:

- Maintain discipline during the lab.
 - Listen and follow the instructions as they are given.
 - Just raise hand if you have any problem.
 - Completing all tasks of each lab is compulsory.
 - Get your lab checked at the end of the session.
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Lab Objective

- ☐ Implementation & understanding of HTTP/HTTPS.
- ☐ Introduction to DNS & configuration of DNS in Cisco Packet Tracer.

Computer Networks
CL3001

LAB - 04
HTTP/HTTPS and DNS

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
KARACHI CAMPUS
FAST SCHOOL OF COMPUTING (AI & DS, CS, CY, SE) Fall
2026

Lab Exercises

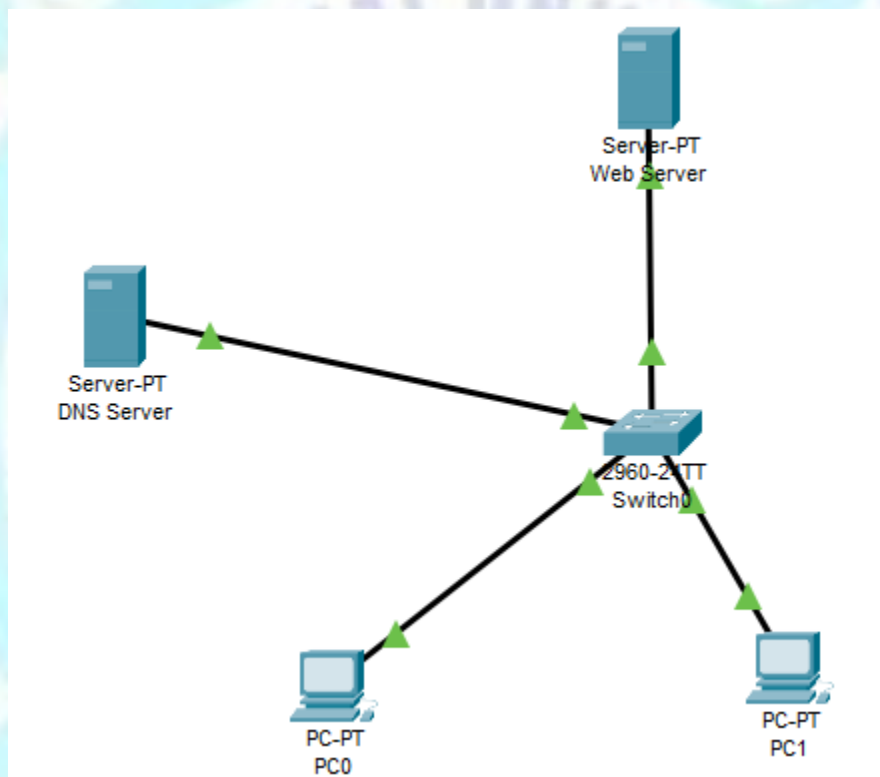
Note:

Server and PC name have roll no.

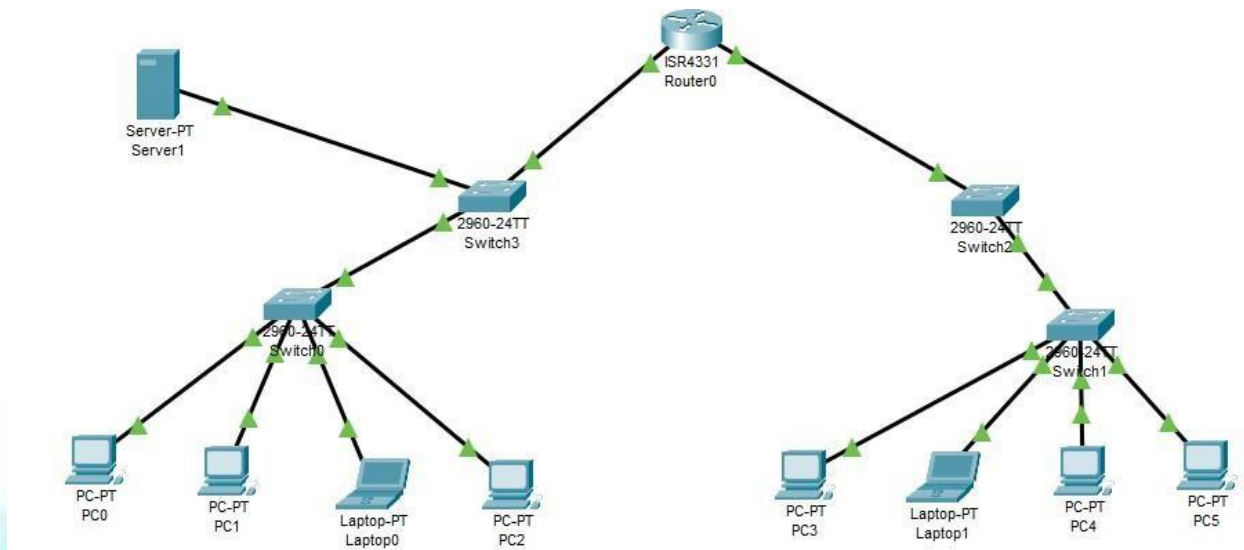
Example:

Server22K0001

1. Create the following network with a DHCP server. Send DNS packets to your network. Attach a snapshot of each step.



2. Implement the given topology. Add some web servers to your network. Implement DNS & add records of your web servers. Attach a snapshot of each step.



3. Which HTTP header sets a "best before" date for cached items?
4. HTTP PUT and POST methods are different or not? Explain the reason in one line.
5. Configure an A record and a CNAME record on your DNS server for your web server, then browse the web server using the domain name instead of its IP address (as done in the manual with "fast-ai"). Attach a snapshot of the DNS query/response in simulation mode and explain in one line what you observe.
6. On your web server, host a page named "cache-test.html" and enable the header that sets a "best before" date for cached items (as discussed in the manual). Browse the page from a PC, go to simulation mode, and capture the HTTP response packet. Attach a snapshot showing the relevant header value.
7. Design a topology with a router connecting two separate networks, with at least two PCs on each side. Assign static IP addresses to all PCs, configure the router interfaces via CLI, and set a default gateway on each PC. Then use the "tracert" command from a PC on one network to a PC on the other network to verify the path through the router. Attach a snapshot of each step along with the trace route output, and explain in one line what it shows.
8. Set up a network with a DNS server and a web server, and assign a domain name to your web server using an A record. Then add an NS record and an SOA record for that domain on the DNS server, as described in the manual. Attach a snapshot of the records added and briefly state, in one line, the purpose of each record.
9. Design a topology with a router connecting two networks, and assign static IP addresses to all PC and router interfaces via CLI. Then run the "show running-config" command via CLI to view the interface configuration. Attach a snapshot of the interface configuration section and identify the IP addresses assigned to each interface.